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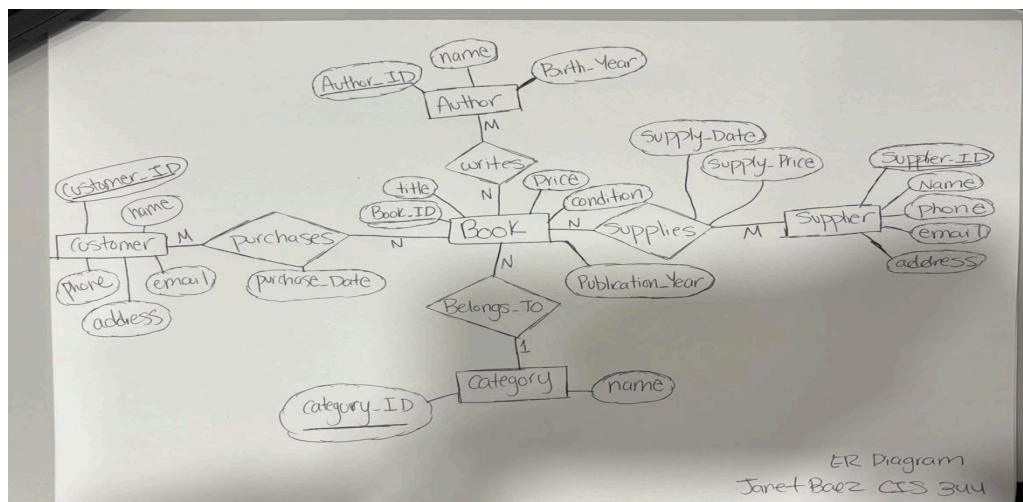
Antique Bookstore Database Project

For my database project I selected an antique bookstore as my mini world because it has different types of information that are connected to each other. The purpose of the database is to help organize important information that an antique bookstore would need to help keep track. The system will store information about things such as books, authors, customers, suppliers and book categories.

To determine the requirements for the antique bookstore database, I considered the type of information that a bookstore would need to organize and manage. I also researched how bookstore inventory is commonly organized and considered what information an employee would need when looking for a book or helping a customer. From this I determined that the database should store information about books, authors, customers, suppliers and multiple categories.

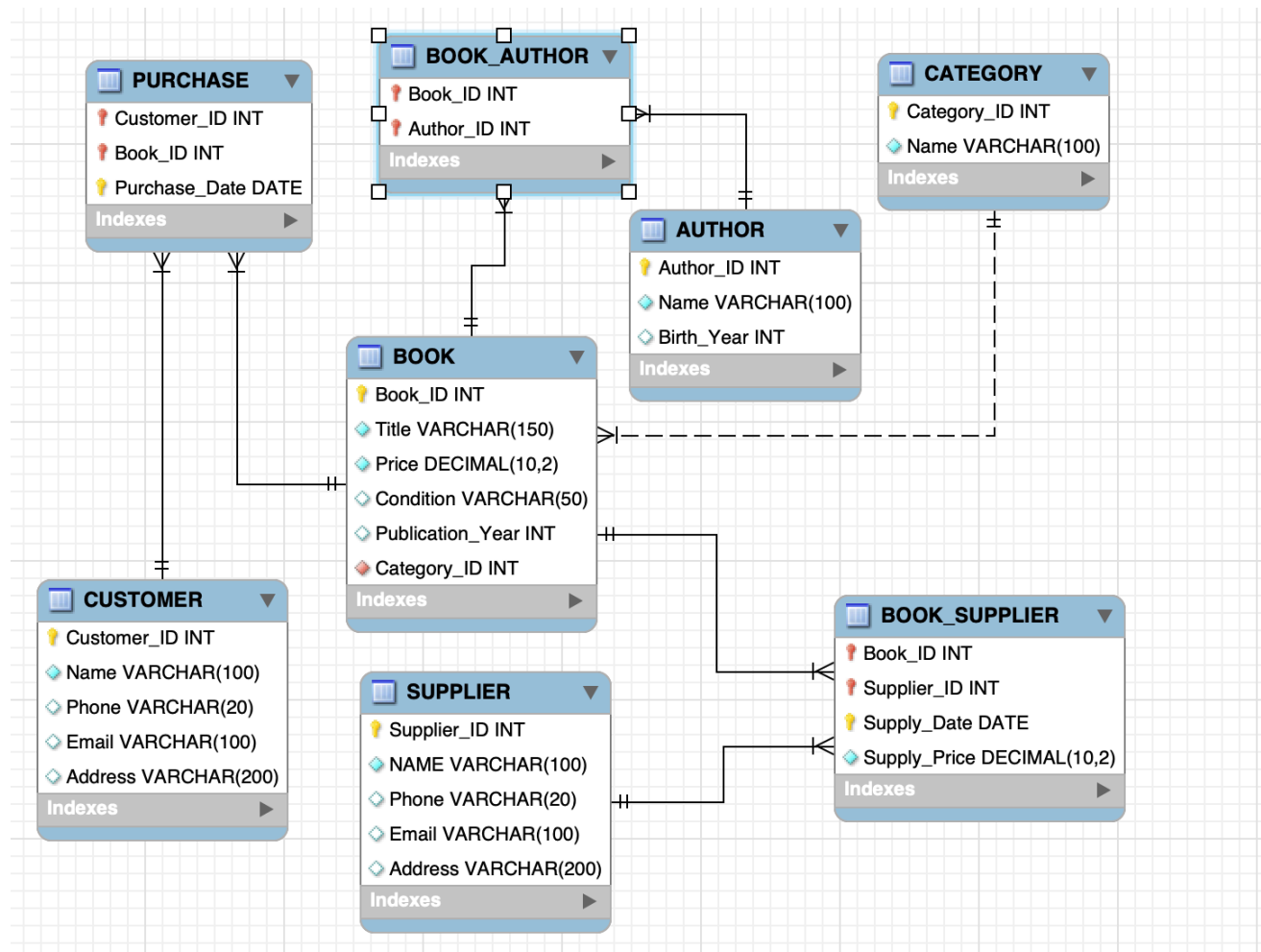
Each book needs identifying information such as its title, price, condition, publication year and author information. Categories are used to organize books into different groups. Supplier information is needed to identify where books are obtained, while customer information allows the bookstore to keep track of basic customer records. Based on these requirements, I selected AUTHOR, BOOK, CUSTOMER, SUPPLIER, and CATEGORY as the main entities for the database. Next, I identified the attributes and relationships needed to connect these entities and used those requirements to create the Chen-style ER diagram.

Chen Style ER Diagram:



The Chen-style ER diagram represents the main entities, attributes, and relationships identified for the antique bookstore database. The diagram uses Chen notation to show how authors, books, customers, suppliers, and categories are related within the system.

UML/EER Diagram



The UML/EER diagram was created using MySQL Workbench. It converts the Chen design into a database design that can be implemented in MySQL.

For the database design, I created both a Chen-style ER diagram and an EER diagram using MySQL Workbench. The Chen diagram was drawn by hand and shows the main entities, attributes, and relationships in the antique bookstore system. The main entities are AUTHOR, BOOK, CUSTOMER, SUPPLIER, and CATEGORY. Junction tables were used to represent the many-to-many relationships between books and authors, books and suppliers and books and customers. Primary keys and foreign keys were used to make sure the tables are connected correctly.

The antique bookstore database was created using MySQL. The database is named antique_bookstore_db and contains eight tables: AUTHOR, BOOK, BOOK_AUTHOR, BOOK_SUPPLIER, CATEGORY, CUSTOMER, PURCHASE, and SUPPLIER. Each main table has a primary key that uniquely identifies its records. Foreign keys connect the related tables and maintain the relationships I created earlier during the design process. Data types such as INT, VARCHAR, DATE, and DECIMAL were selected depending on the information being stored. After creating the database, I used the “show tables” command to verify that all eight tables were successfully created.

Challenges:

One challenge I experienced during the project was converting the many-to-many relationships from the Chen diagram into the relational database model. These relationships required junction tables so that they could be properly implemented in MySQL. I also experienced a connection issue while attempting to forward engineer the model from MySQL Workbench. To solve this problem, I used my working MySQL connection and executed the SQL script directly. This allowed me to successfully create and verify the database while keeping the same design represented in my EER model.

Final:

The final result is a working relational database for an antique bookstore. The database can organize information about books, authors, customers, suppliers, categories, purchases and book suppliers. The finished design includes primary keys, foreign keys and relationships that connect the different parts of the system. The SQL script executed beautifully and all eight required tables were created in MySQL.